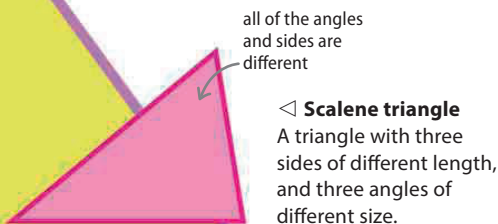
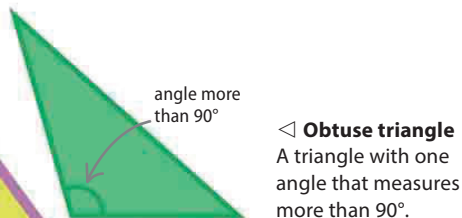
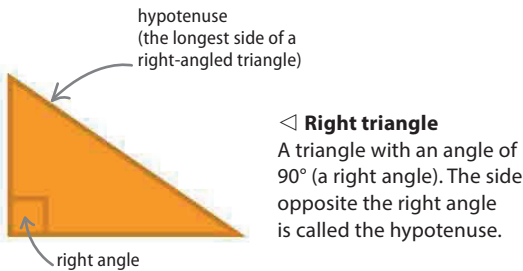
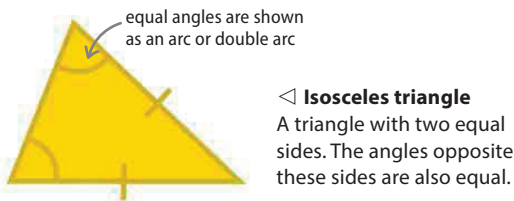
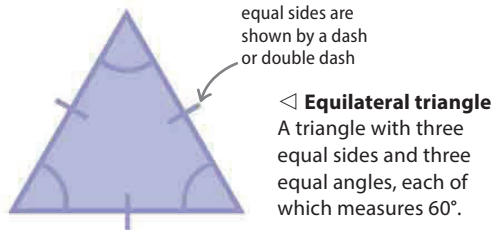


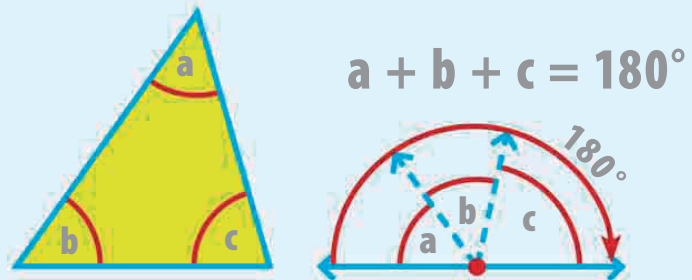
Types of triangles

There are several types of triangles, each with specific features, or properties. A triangle is classified according to the length of its sides or the size of its angles.



Interior angles of a triangle

A triangle has three interior angles at the points where each side meets. These angles always add up to 180° . If rearranged and placed together on a straight line, the interior angles would still add up to 180° , because a straight angle always measures 180° .

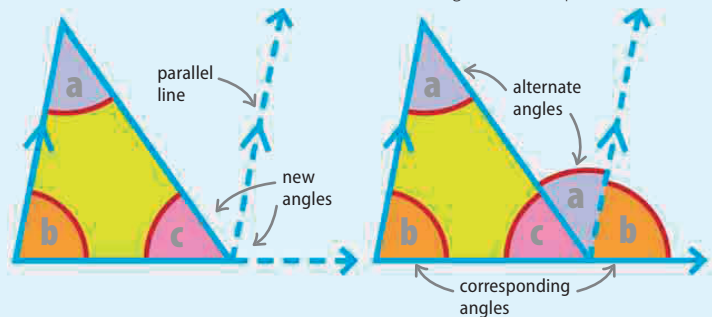


Proving that the angle sum of a triangle is 180°

Adding a parallel line produces two types of relationships between angles that help prove that the interior sum of a triangle is 180° .

Draw a triangle, then add a line parallel to one side of the triangle, starting at its base, to create two new angles.

Corresponding angles are equal and alternate angles are equal; angles c , a , and b sit on a straight line so together add up to 180° .



Exterior angles of a triangle

In addition to having three interior angles a triangle also has three exterior angles. Exterior angles are found by extending each side of a triangle. The exterior angles of any triangle add up to 360° .

$$x + y + z = 360^\circ$$

each exterior angle of a triangle is equal to the sum of the two opposite interior angles, so $y = p + q$

